

Data Science for Beginners

AI Learning Hub - Comprehensive Course Guide

Generated for educational purposes

Data Science Workflow

Data Science follows a structured workflow:

1. Problem Definition: Clearly define the business question.
2. Data Collection: APIs, databases, web scraping, surveys.
3. Data Cleaning: Handle missing values, outliers, duplicates.
4. Exploratory Data Analysis (EDA): Visualize distributions, correlations.
5. Feature Engineering: Create meaningful features from raw data.
6. Modeling: Select and train appropriate algorithms.
7. Evaluation: Assess model performance on test data.
8. Deployment: Serve model via APIs (Flask, FastAPI).

Tools: Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Jupyter.

Statistics for Data Science

Essential statistics concepts:

Descriptive Statistics: Mean, Median, Mode, Standard Deviation, Variance, Percentiles, IQR (Interquartile Range).

Probability Distributions: Normal (Gaussian), Binomial, Poisson, Uniform.

Central Limit Theorem: Sample means approach normal distribution.

Hypothesis Testing:

- Null Hypothesis (H_0) vs Alternative Hypothesis (H_1).
- p-value: probability of observing data given H_0 is true.
- Significance level (α): typically 0.05.
- Type I Error (false positive), Type II Error (false negative).
- t-test, chi-squared test, ANOVA for different scenarios.

Data Visualization

Effective visualization communicates insights clearly.

Chart Types and When to Use Them:

- Bar Charts: Comparing categories.
- Line Charts: Trends over time.
- Scatter Plots: Relationship between two variables.
- Histograms: Distribution of a single variable.
- Box Plots: Distribution summary with outliers.
- Heatmaps: Correlation matrices, confusion matrices.
- Pair Plots: All pairwise relationships in a dataset.

Libraries: Matplotlib (low-level control), Seaborn (statistical plots), Plotly (interactive), Altair (declarative). Best practices: label axes, use appropriate scales, avoid chartjunk, tell a story with data.